

FACULTY OF ENGINEERING AND BASIC SCIENCES
ACADEMIC PROGRAM: DATA ENGINEERING AND ARTIFICIAL INTELLIGENCE

COURSE: ETL (G01)

ETL PROJECT — DATA ENGINEERING FOR SUSTAINABLE DEVELOPMENT IN COLOMBIA
PHASE 1: FROM ANALYTICAL REQUIREMENTS TO A DIMENSIONAL DATA WAREHOUSE

1. OBJECTIVE

Design and implement the first stage of an analytical data solution to study a specific Colombian problem associated with a Sustainable Development Goal (SDG). The project must demonstrate coherence across problem understanding, analytical requirements, dataset selection and assessment, data profiling, data preparation, dimensional modeling, ETL, the Data Warehouse, SQL/KPIs, Business Intelligence, and the interpretation of findings for decision support.

Guiding perspective: Problem → Requirements → Data → ETL → Dimensional Model → Data Warehouse → KPIs → BI → Decision Support

This is an academic first implementation. The emphasis is not merely on programming scripts, but on designing a consistent analytical solution whose technical components respond to a clearly defined need.

2. COLOMBIAN PROBLEM DEFINITION

Each team must select one SDG and identify a specific Colombian problem related to it. The SDG provides the general framework; the concrete Colombian problem is the object of analysis. Delimit the problem geographically, sectorally, or by population whenever appropriate. For example:

Weak formulation	Improved formulation
Analyze SDG 3 in Colombia.	Analyze disparities in access to healthcare services across Colombian departments.

The problem definition must include:

1. Selected SDG.
2. Relevant SDG target(s), when applicable.
3. Colombian context and evidence that motivates the problem.
4. Geographic scope.
5. Population, sector, or phenomenon of interest.
6. A precise problem statement.
7. Potential stakeholders or users of the analytical solution.
8. Relevance of the analysis for decision-making.

3. ANALYTICAL OBJECTIVE AND REQUIREMENTS

Define one analytical/business objective and at least five analytical requirements, identified as R1–R5. Each requirement must represent a genuine analytical need, relate directly to the Colombian problem, be answerable with the selected dataset, support a decision or insight, and require analysis rather than simple record retrieval.

Important distinction: A chart, KPI, SQL query, or dashboard is not an analytical requirement. These are analytical outputs used to satisfy a requirement.

Requirements Matrix (R1–R5)

ID	Analytical Requirement	Business Question	Decision Supported / Insight
R1			
R2			
R3			
R4			
R5			

4. SDG ALIGNMENT

Briefly justify how the selected Colombian problem and the analytical requirements relate to the chosen SDG. When pertinent, identify associated SDG targets or indicators.

Scope: The project does not need to reproduce an official SDG indicator. However, the relationship between the selected problem and the SDG must be clearly justified.

5. DATA SOURCE SELECTION

Use one primary data source for this delivery. The dataset must relate directly to the selected Colombian problem and should preferably come from an official Colombian government or public institution, such as Datos Abiertos Colombia, DANE, DNP, a ministry, a national agency, or a departmental or municipal open-data portal. Other sources may be considered when their relevance and reliability are justified.

The following characteristics should be used as general dataset-selection guidelines rather than as inflexible thresholds:

- Approximately 10,000 records or more.
- At least 8–10 relevant attributes.
- Sufficient analytical richness to support R1–R5.
- Temporal information whenever applicable.
- Categorical or descriptive attributes that can support meaningful dimensions.
- Information suitable for facts, measures, or KPIs.
- A clearly identifiable source, data owner, and dataset description.

Possible exception: Smaller datasets may be accepted when their analytical value, geographic coverage, temporal depth, or relevance to the selected problem justifies their use.

6. DATASET SUITABILITY ASSESSMENT

Assess the dataset before implementation and document the following:

Dataset Suitability Matrix

Criterion	Assessment
Institution / Data Owner	
Source URL / Access Mechanism	
Format	
Number of Records	
Number of Attributes	
Geographic Coverage	
Temporal Coverage	
Relevant Numerical Measures	
Relevant Categorical Attributes	
Potential Data-Quality Issues	
Relationship with Analytical Requirements	
Suitability for Dimensional Modeling	
Unit of Observation in the Source	

7. DATA PROFILING AND QUALITY ASSESSMENT

Perform and document an initial data profile that includes, at minimum:

Number of records and attributes.

- Column names and data types.
- Missing values and duplicate records.
- Unique values and cardinality.
- Date coverage.
- Geographic coverage, when applicable.
- Invalid, inconsistent, or suspicious values.
- Relevant numeric ranges and basic descriptive statistics.
- Issues that could affect the requirements or the dimensional model.

Purpose: The objective is to understand the data and identify relevant quality issues.

8. REQUIREMENTS-TO-DATA TRACEABILITY

Show that the selected data can support each analytical requirement and identify the transformations needed to make that support possible.

Requirement	Required Attributes	Transformation Needed	Expected KPI / Analysis
R1			
R2			
R3			
R4			
R5			

Design principle: Do not start by designing tables. Start by understanding what the problem requires the analytical system to answer.

9. DATA PREPARATION STRATEGY

Define a justified preparation strategy based on the profiling findings and the analytical requirements. It may include:

- Data-type conversion.
- Missing-value handling.
- Duplicate analysis and treatment.
- Date parsing and standardization.
- Categorical harmonization.
- Identifier and text standardization.
- Handling invalid or inconsistent values.
- Creation of derived attributes only when justified by the requirements.

Not every detected issue must necessarily be modified. Each preparation decision must be justified according to its impact on the analytical requirements and dimensional model.

10. DECLARE THE GRAIN

The declared grain must describe the business event, measurement, or observation represented by each row of the Fact Table.

Declare the grain before designing dimensions and facts. It must be compatible with both the analytical requirements and the available source data.

Required statement: One row in Fact_____ represents _____.

Mandatory condition: No dimensional model will be accepted without an explicitly declared grain.

11. DIMENSIONAL DATA MODEL

Design a dimensional model that includes:

- At least one Fact Table.

- The Dimension Tables required by the declared grain and R1–R5.
- Surrogate keys.
- Explicit primary-key and foreign-key relationships.
- Measures and descriptive attributes.
- A Star Schema diagram.

Analytical justification: Every dimension, fact, measure, and relevant attribute must have an analytical justification. Do not create dimensions simply because categorical columns exist in the source dataset.

12. REQUIREMENTS-TO-MODEL VALIDATION

Requirement	Dimension(s)	Measure(s)	Expected Query/KPI	Supported ?
R1				Yes / No
R2				Yes / No
R3				Yes / No
R4				Yes / No
R5				Yes / No

Validation rule: If a requirement cannot be answered by the proposed model, revise the model before implementation.

13. ETL PIPELINE

Implement a reproducible Python ETL pipeline with clearly separated responsibilities:

Extract

- Read or acquire the source.
- Preserve the raw data.
- Document acquisition.
- Do not perform business transformations during extraction.

Transform — Data Preparation

- Clean and standardize the data.
- Correct data types.
- Prepare dates, categories, and identifiers.
- Create derived attributes only when justified.

Transform — Dimensional Transformation

- Create dimensions and remove duplicated dimension members where necessary.
- Generate surrogate keys and map source identifiers to them.
- Create the Fact Table according to the declared grain.
- Include only facts and relationships justified by the requirements.

Validate

- Check row counts, critical nulls, uniqueness, PK/FK constraints, and referential integrity.
- Check relevant ranges and business rules.
- Perform basic reconciliation with the source.

Load

- Create the Data Warehouse schema.
- Load dimensions first and then the Fact Table.
- Verify referential integrity after loading.

14. DATABASE TECHNOLOGY

Required technology: Implement the Data Warehouse using PostgreSQL or MySQL. Other database technologies are not permitted unless previously approved.

15. ANALYTICAL QUERIES AND KPIS

Implement at least one analytical SQL query for each requirement R1–R5 and define the corresponding metric or KPI when applicable, and summarize the main result.

Requirement	Analytical Query	Metric or KPI	DW Used	Tables	Main Result
R1					
R2					
R3					
R4					
R5					

Mandatory source: All analytical queries must use the Data Warehouse. Queries against the raw source file or intermediate DataFrames do not satisfy this requirement.

16. BUSINESS INTELLIGENCE

Build an analytical dashboard or BI report in Power BI, Tableau, Looker Studio when a viable database connection is implemented, or another BI tool approved by the instructor. The BI solution must connect to the PostgreSQL or MySQL Data Warehouse and must not read the source file directly.

When applicable, the dashboard should include:

- Temporal analysis.
- Geographic or territorial analysis.
- Comparative analysis.
- Relevant KPIs and measures.
- Useful filters.

BI condition: Python plotting libraries do not satisfy the BI requirement.

17. ANALYTICAL INTERPRETATION

Document at least three relevant findings. For each finding, explain:

- What does the data show?
- Which requirement does it address?
- Why is the finding relevant in the Colombian context?
- What decision or further investigation could it support?

18. TECHNOLOGIES

Use the following technologies:

- Python and Pandas.
- Jupyter Notebook for initial profiling.
- SQL.
- PostgreSQL or MySQL.
- Power BI, Tableau, Looker Studio, or another approved BI tool.
- Git and GitHub.

This project follows an ETL approach: the required transformations must occur before the analytical model is loaded into the Data Warehouse.

19. SYSTEM ARCHITECTURE

Design a high-level architecture diagram showing the main components and data flow of the proposed analytical solution. For example:

Source Dataset → Extraction → Data Profiling/Preparation → Dimensional Transformation → PostgreSQL/MySQL Data Warehouse → SQL/KPIs → BI → Business Insights

The architecture should represent components and data flow, not implementation code.

20. GITHUB REPOSITORY

A recommended repository structure is shown below. Equivalent structures are acceptable when responsibilities remain clear.

```

etl-project-first-delivery/
├── data/
│   ├── raw/
│   └── processed/
├── notebooks/
│   └── data_profiling.ipynb
├── src/
│   ├── extract.py
│   ├── transform.py
│   ├── validate.py
│   ├── dimensional_model.py
│   ├── load.py
│   └── main.py
├── sql/
│   ├── create_dw.sql
│   └── analytical_queries.sql
├── docs/
│   ├── architecture.png
│   ├── star_schema.png
│   └── dashboard.png
├── README.md
├── requirements.txt
└── .gitignore
    
```

Large dataset: Provide the exact source URL and reproducible acquisition instructions instead of forcing the raw dataset into the repository.

21. README

The README must function as the project's technical documentation and include:

- Project title, selected SDG, and Colombian problem.
- Analytical objective, stakeholders, and R1–R5.
- SDG alignment.
- Dataset source, owner, link, data acquisition instructions, and suitability assessment.
- Requirements-to-data traceability and profiling findings.
- Data preparation strategy.
- System architecture, business process, and grain.
- Star Schema and justification of dimensions and facts.
- ETL process and validation rules.
- Data Warehouse implementation and reproduction/run instructions.
- Analytical queries/KPIs and BI dashboard evidence.
- Main findings, limitations and assumptions.
- Team members and responsibilities.

21. DELIVERABLE

Submission: Submit ONLY the GitHub repository URL through the LMS. All evidence and documentation must be organized within the repository; no separate PDF report is required.

22. PRESENTATION AND DEFENSE

The team will present and defend the implemented analytical solution. Every member must understand the problem, requirements, dataset, profiling, grain, dimensional model, ETL, SQL, Data Warehouse, BI solution, and findings.

Individual mastery: During the presentation, any team member may be asked to explain any component of the implemented solution.

Demonstration: The instructor may request execution or demonstration of selected components of the implemented solution.

23. EVALUATION

The final grade for this delivery is distributed as follows:

GitHub Repository Criterion	Weight
Colombian problem, SDG alignment, objective, and R1–R5	10%
Dataset selection, suitability assessment, and profiling	10%
Requirements-to-data and requirements-to-model traceability	5%
Grain and dimensional modeling	15%
ETL implementation and validation	15%
PostgreSQL/MySQL Data Warehouse implementation	8%
SQL queries and KPIs	5%
BI dashboard and analytical interpretation	10%
Reproducibility, repository organization, and documentation	2%
GitHub Repository subtotal	80%

Presentation and Defense Criterion	Weight
Understanding of the problem and the complete solution	5%
Justification of analytical and technical decisions	4%
Interpretation of findings	4%
Response to questions and individual mastery	4%
Professional communication	3%
Presentation and Defense subtotal	20%

Total: GitHub Repository (80%) + Presentation and Defense (20%) = 100%.